ECG Chamber Hypertrophy — RAG Knowledge Source

English • Text only • No tables • Condensed from AHA/ACCF/HRS 2009 Part V

Source identity. Hancock EW, Deal BJ, Mirvis DM, Okin P, Kligfield P, Gettes LS. AHA/ACCF/HRS Recommendations for the Standardization and Interpretation of the Electrocardiogram, Part V: Electrocardiogram Changes Associated With Cardiac Chamber Hypertrophy. Circulation. 2009;119:e251–e261. DOI: 10.1161/CIRCULATIONAHA.108.191097.

Scope. This source concerns ECG changes associated with cardiac chamber hypertrophy. It is not a general ECG interpretation guide and should not be treated as one.

RAG purpose. The document retains definitions, major ECG patterns, commonly cited voltage criteria, and limitations that are useful for retrieval. Reference lists and author/contributor information are excluded.

1. General Principles

ECG is used clinically to detect and assess cardiac chamber hypertrophy.

The principal ECG changes associated with ventricular hypertrophy include increased QRS amplitude and duration, changes in QRS vectors, ST-segment and T-wave abnormalities, and P-wave abnormalities.

ECG criteria for hypertrophy have historically been compared with clinical, autopsy, echocardiographic, CT, and magnetic resonance assessments of ventricular mass.

ECG remains clinically useful because it is widely available, convenient, and inexpensive.

No ECG criterion should be interpreted as an absolute measurement of anatomic chamber mass.

2. Left Ventricular Hypertrophy — LVH

Most commonly used ECG criteria for LVH are based on QRS voltage.

The Sokolow–Lyon criterion uses the sum of S-wave amplitude in V1 and R-wave amplitude in V5 or V6. A value of 35 mm is a commonly cited threshold in the source.

The Cornell voltage criterion uses the sum of S-wave amplitude in V3 and R-wave amplitude in aVL. The source lists thresholds of 28 mm in men and 20 mm in women.

Other voltage criteria include R in aVL ≥11 mm, R in aVF ≥20 mm, and S in V1 ≥23 mm.

QRS duration may be increased in LVH. This can appear as diffuse QRS prolongation or increased time from QRS onset to the R-wave peak in V5 or V6.

Voltage-duration products and other composite criteria have also been developed.

3. LVH Diagnostic Performance

Different LVH criteria do not identify exactly the same patients.

Sensitivity of individual ECG criteria for LVH is generally low, often below 50%, while specificity is generally high, often around 85–90%.

Diagnostic performance varies with the criterion and the patient population.

Interpretations should specify which criteria were used.

Automated ECG systems should consider multiple validated criteria rather than relying on a single voltage measurement.

QRS voltage can be influenced by age, sex, body habitus, and population characteristics.

4. Right Ventricular Hypertrophy — RVH

RVH may produce increased R-wave amplitude in right precordial leads, increased S-wave amplitude in left precordial leads, rightward QRS axis, and other QRS changes.

The source lists a commonly cited criterion of R in V1 + S in V5 or V6 ≥10.5 mm.

An R-wave peak in V1 with QRS duration ≥0.12 seconds and a QR pattern in V1 are among described criteria or supporting findings.

Supporting findings can include an RSR pattern in V1 with QRS duration ≥0.12 seconds, rightward axis patterns, and T-wave abnormalities in right precordial leads.

No single criterion or limited set of criteria should be used exclusively for RVH diagnosis.

RVH probability should be interpreted in light of clinical diagnoses such as congenital heart disease, valvular disease, or chronic pulmonary disease.

RVH criteria should be considered in relation to age, sex, body habitus, and other patient characteristics.

5. Biventricular Hypertrophy

ECG detection of biventricular hypertrophy has particularly low sensitivity because increased QRS vectors from RVH and LVH can partially cancel each other.

Biventricular hypertrophy should be suggested when accepted criteria for both RVH and LVH are present.

Right axis deviation in a patient with ECG evidence of LVH, prominent S waves in V5 or V6, tall biphasic R/S complexes, and signs of right atrial abnormality can support suspicion of biventricular hypertrophy.

Because sensitivity is low, absence of ECG criteria does not exclude biventricular hypertrophy.

6. Left Atrial Abnormality

Abnormal P-wave patterns can be associated with left or right atrial abnormalities.

A total P-wave duration of 120 ms or more is commonly present in patients with ECG signs considered to represent left atrial abnormality.

A purely negative P wave in V1 can be suggestive but is not specific by itself.

Left atrial abnormality is associated with changes in the later P-wave forces and may be accompanied by P-wave axis changes.

7. Right Atrial Abnormality

Right atrial abnormality is typically associated with increased P-wave amplitude and a tendency toward rightward P-wave vector shift.

A tall upright P wave in lead II greater than 2.5 mm is a characteristic finding.

Prominent initial positivity of the P wave in V1 or V2 of 1.5 mm or more also indicates right atrial abnormality.

Rightward P-wave axis and a peaked P-wave form can provide supporting evidence.

Total P-wave duration is usually normal in right atrial abnormality, although important exceptions occur in some patients with repaired congenital heart disease.

8. Pediatric Considerations

Pediatric ECG voltage criteria for ventricular hypertrophy are age-dependent.

The source provides separate age-related criteria for pediatric LVH and RVH.

Adult voltage thresholds should not be automatically applied to children.

When pediatric criteria are used, age and the appropriate pediatric reference values must be retained as metadata.

9. Clinical Interpretation Rules

ECG findings associated with chamber hypertrophy should be interpreted together rather than as isolated abnormalities.

A positive voltage criterion does not by itself establish anatomic hypertrophy.

A negative ECG does not reliably exclude hypertrophy because ECG sensitivity is limited.

The clinical context and, when needed, imaging should be considered when confirming chamber hypertrophy.

Criteria may behave differently across patient populations and body habitus.

10. RAG Retrieval Rules

Use one ECG concept per chunk.

Keep each voltage formula and its threshold in the same chunk.

Never separate a threshold from the lead configuration to which it applies.

Keep adult and pediatric criteria in separate chunks.

Keep LVH, RVH, biventricular hypertrophy, left atrial abnormality, and right atrial abnormality separate.

Preserve limitations such as low sensitivity and population dependence because they prevent overconfident retrieval-based answers.

Do not ingest the paper's reference list or author/contributor disclosures as clinical knowledge.

Recommended metadata: source_id=AHA_ECG_HYPERTROPHY_2009; topic; chamber; criterion_type; age_group; keywords; original page if available.

11. Example Retrieval Units

Topic: Sokolow–Lyon LVH criterion.

Question: What is the Sokolow–Lyon voltage criterion for LVH?

Answer: It uses the sum of S-wave amplitude in V1 and R-wave amplitude in V5 or V6; the source lists 35 mm as a commonly cited threshold.

Source ID: AHA_ECG_HYPERTROPHY_2009.

Keywords: LVH, left ventricular hypertrophy, Sokolow-Lyon, V1, V5, V6, QRS voltage.

Topic: RVH limitation.

Question: Can one ECG criterion diagnose RVH reliably?

Answer: No. The source recommends against relying exclusively on a single criterion or limited set of criteria; clinical context and validated criteria should be considered.

Source ID: AHA_ECG_HYPERTROPHY_2009.